

Exact Reduction Certificates for Laurent Periods

The G, U, V, J, A, Ξ formalism

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Public draft with one worked example

Abstract

We consider Laurent polynomials $F \in \mathbb{Q}[x^{\pm 1}, y^{\pm 1}]$ and their constant-term power series

$$\Phi(t) = \left[\frac{1}{1 - tF} \right]_0 = \sum_{n \geq 0} [F^n]_0 t^n.$$

A finite exact reduction constructs a differential annihilator A for Φ together with an explicit certificate $\Xi = (\Xi_x, \Xi_y)$. The general symbolic implementation uses the G, U, V, J formalism developed in Chapter 3 of Bradley Klee's dissertation. The coefficient matrix G records the reduction identity, its solution gives the reduced part U and certificate data V , and J records the derivative correction made when the pole order is lowered. Linear dependence among the resulting finite-dimensional data gives A , while the accumulated V -data give Ξ without a second solve. The worked public example is the cubic Hamiltonian recorded with OEIS A303790; its factorization gives a shorter Laurent telescoper, paired with an independent Hamiltonian certificate. The accompanying exact data reproduce its coefficient sequence, differential equation, recurrence, and both certificate checks.

1 Formal reductions

Put $K = \mathbb{Q}(t)$ and

$$\rho = 1 - tF, \quad D_x = x \frac{\partial}{\partial x}, \quad D_y = y \frac{\partial}{\partial y}.$$

The coefficient of $x^0 y^0$ in either $D_x(R)$ or $D_y(R)$ is zero. Thus a pair of rational functions $\Xi = (\Xi_x, \Xi_y)$ satisfying

$$A\left(\frac{1}{\rho}\right) = D_x(\Xi_x) + D_y(\Xi_y) \tag{1}$$

proves $A\Phi = 0$.

Fix a positive integer r . Let w, a, b_x, b_y be Laurent polynomials. If

$$w = \rho a - D_x(\rho)b_x - D_y(\rho)b_y, \tag{2}$$

then the product rule gives the reduction identity

$$\frac{w}{\rho^{r+1}} = \frac{a - \frac{1}{r}(D_x(b_x) + D_y(b_y))}{\rho^r} + D_x\left(\frac{b_x}{r\rho^r}\right) + D_y\left(\frac{b_y}{r\rho^r}\right). \tag{3}$$

Thus lowering the pole order of w/ρ^{r+1} reduces to solving (2) for (a, b_x, b_y) . The relevant linear map is

$$(a, b_x, b_y) \mapsto \rho a - D_x(\rho)b_x - D_y(\rho)b_y. \tag{4}$$

Let $B = (b_1, \dots, b_d)$ be the ordered Laurent-monomial basis for each of a, b_x, b_y . Let $C = (c_1, \dots, c_m)$ be an ordered Laurent-monomial basis large enough to contain $B, FB, D_x(F)B$, and $D_y(F)B$. For a coefficient column $q = (q_1, \dots, q_d)^T \in K^d$, write

$$\langle q \rangle = \sum_{j=1}^d q_j b_j.$$

Angle brackets always reconstruct a Laurent polynomial in the basis B . The rows of G are indexed by the basis C , and its columns are indexed by three ordered copies of B , corresponding to a, b_x, b_y .

The matrix

$$E \in K^{m \times d}$$

is the inclusion of the B -space into the C -space. Its j th column is b_j written in the basis C .

The matrix

$$G \in K^{m \times 3d}$$

represents the map (4). We compute

$$U, V_x, V_y \in K^{d \times d}$$

from the exact matrix identity

$$G \begin{pmatrix} U \\ V_x \\ V_y \end{pmatrix} = E. \tag{5}$$

The stacked matrix has size $3d \times d$, so both sides of (5) have size $m \times d$.

For any $q \in K^d$, multiplication of (5) by q gives

$$G \begin{pmatrix} Uq \\ V_x q \\ V_y q \end{pmatrix} = Eq.$$

Therefore the Laurent polynomial $w = \langle q \rangle$ is split by (2) with

$$a = \langle Uq \rangle, \quad b_x = \langle V_x q \rangle, \quad b_y = \langle V_y q \rangle.$$

If G is square and nonsingular, then

$$\begin{pmatrix} U \\ V_x \\ V_y \end{pmatrix} = G^{-1} E.$$

If G is rectangular, exact Gauss–Jordan elimination is applied to the augmented system. Free variables are set to zero, and the resulting solution is accepted only when substitution gives zero residual in (5).

Put

$$V = \begin{pmatrix} V_x \\ V_y \end{pmatrix} \in K^{2d \times d}.$$

The matrix

$$J \in K^{d \times 2d}$$

represents differentiation of the certificate numerators:

$$\left\langle J \begin{pmatrix} q_x \\ q_y \end{pmatrix} \right\rangle = D_x \langle q_x \rangle + D_y \langle q_y \rangle.$$

Substitution in (3) gives the one-step reduction matrix

$$\text{Lower}_r = U - \frac{1}{r}JV \in K^{d \times d}. \quad (6)$$

A kernel relation among reduced coefficient columns gives A ; the same linear combination of the stored V_x, V_y columns gives Ξ .

2 Example

The cubic Hamiltonian and OEIS A303790.

Let

$$K(p, q) = p^2 + q^2 + p^3 + q^3, \quad t = E/32. \quad (7)$$

The normalized period at either finite center has the form

$$\Pi(E) = \sum_{n \geq 0} A_n (E/32)^n.$$

The equality of the two local coefficient sequences follows from the affine involution $(p, q) \mapsto (-p - 2/3, -q - 2/3)$, which sends K to $8/27 - K$.

Define the integral Laurent polynomial

$$F_{303790}(w, y) = \frac{(w+1)^3(y+1)^2(y^2-4y+1)^2}{w^2y^3}. \quad (8)$$

Lagrange inversion for the radial equation gives the exact coefficient identity

$$A_n = [F_{303790}(w, y)^n]_0. \quad (9)$$

Thus integrality is immediate.

For the exact recurrence proof, factor $F_{303790} = B(w)C(y)$. The w -factor gives $[B^n]_0 = \binom{3n}{2n}$. For the y -factor, the supplied rational function $R(n, y)$ satisfies

$$P_0(n) + P_1(n)C + P_2(n)C^2 = y \frac{\partial R}{\partial y} + n y \frac{C'}{C} R. \quad (10)$$

After multiplication by C^n , the right side is $y \partial_y (RC^n)$ and has zero constant term. Combining the resulting recurrence for $[C^n]_0$ with the exact ratio for $\binom{3n}{2n}$ gives

$$(n+1)^2 A_{n+1} - 12(27n^2 + 27n + 5)A_n + 2592(3n-2)(3n+2)A_{n-1} = 0. \quad (11)$$

The full P_0, P_1, P_2, R data are kept in the accompanying exact certificate.

For $Y(t) = \sum_{n \geq 0} A_n t^n$, this is equivalent to

$$\begin{aligned} t(108t-1)(216t-1)Y'' + (69984t^2 - 648t + 1)Y' \\ + 60(216t-1)Y = 0. \end{aligned} \quad (12)$$

The independent Hamiltonian certificate proves

$$E(27E-8)(27E-4)\Pi'' + (2187E^2 - 648E + 32)\Pi' + 15(27E-4)\Pi = 0.$$

Substitution of $E = 32t$ gives exactly (12). Both certificate residuals are zero by exact coefficient reduction. The first values are

$$1, 60, 7380, 1090320, 176978340, 30471320880, \dots,$$

in agreement with OEIS A303790.

3 General G, U, V, J pseudocode

The public A303790 packet also contains the shorter factorized replay used in the worked example. The following is the general matrix-reduction route.

1. Read F as an exact Laurent polynomial; set $\rho = 1 - tF$ and list the exponent pairs occurring in 1 and F .
2. Form every pairwise sum of those exponent pairs; let B be all integer points in their convex hull, ordered lexicographically.
3. Let C be the union of B with every translate $B + (i, j)$ for which $x^i y^j$ occurs in F ; order C lexicographically.
4. Represent Laurent polynomials on B or C by exact coefficient dictionaries keyed by exponent pairs.
5. For each b_j in B , place the C -coordinates of ρb_j , $-D_x(\rho)b_j$, and $-D_y(\rho)b_j$ in columns j , $d + j$, and $2d + j$ of $G \in K^{m \times 3d}$.
6. Build $E \in K^{m \times d}$ and $J \in K^{d \times 2d}$ in the ordered bases B, C .
7. Solve $G(U, V_x, V_y)^T = E$ by exact Gauss–Jordan elimination, or solve only $G(u, v_x, v_y)^T = E f_2$ when only the F^2 column is required; then impose $\text{Lower}_2 f_2 = \alpha r + \beta f$.
8. Evaluate this system at the first regular integer value of t ; perform exact Gauss–Jordan elimination.
9. At successive regular integer values of t , solve that same square block exactly over $\mathbb{Q}$; keep training values and three held-out values.
10. For each unknown coefficient, reconstruct the rational function of least total numerator-plus-denominator degree fitting the training values; reject it unless it also fits every held-out value.
11. Set all free unknowns to zero, rebuild the symbolic solution, and substitute it into the original block system over $\mathbb{Q}(t)$.
12. Clear stated denominators, expand in the declared monomial basis, and require every coefficient of every residual to be zero.
13. Extract the selected column vectors u, v_x, v_y and α, β ; verify $G(u, v_x, v_y)^T = E f_2$ and $\text{Lower}_2 f_2 = \alpha r + \beta f$.
14. Form $X = \begin{pmatrix} 1 & 0 & 2\alpha \\ 0 & 1 & 2\beta \end{pmatrix}$; compute a nonzero kernel vector and normalize it by denominator clearing, polynomial-gcd removal, integer-content removal, and sign fixing.
15. Write the normalized vector as $(p_0, p_1, p_2)^T$; set $A = p_0 + p_1 \partial_t + p_2 \partial_t^2$ and $\Xi = (p_2 \langle v_x \rangle / \rho^2, p_2 \langle v_y \rangle / \rho^2)$.
16. Verify $Xp = 0$ and verify the cleared coefficient identity corresponding to $A(1/\rho) = D_x(\Xi_x) + D_y(\Xi_y)$.
17. Convert A to $\sum_k t^k Q_k(\theta)$ with $\theta = t \partial_t$; output $\sum_k Q_k(n - k)[F^{n-k}]_0 = 0$.
18. Generate the requested constant terms by lattice convolution; test the recurrence on all available indices and compare the first five values with direct power expansion.
19. Serialize the ordered bases, sparse matrices, reduction data, A , Ξ , recurrence, constant terms, and exact pass/fail checks; verify all declared dimensions and nonzero counts.
20. Report elapsed time after basis construction, matrix construction, the exact solve, reduction replay, certificate replay, term generation, and serialization; stop at the first failed check.
21. Do not call a generic simplification routine. Use only named operations whose effects are visible in the proof record.

4 Algorithm and data model

The public package contains the exact A303790 replay, the eleven-record public catalogue, the catalogue verifier, the generic G, U, V, J implementation, and its JSON Schema. The Python source and schema remain separate files rather than typeset appendices. This PDF explains the calculation; the package files supply the literal executable identities and coefficient data. No private reference record is copied into the public artifact.

The public replay verifies every stored coefficient list and differential equation, then repeats both exact certificates for each completed catalogue entry. The A303790 scalar and Laurent certificates are also replayed by their independent scripts. The generic G, U, V, J verifier takes an explicit data root, rebuilds G and J from the stated Laurent polynomial, and writes its report beside the selected data rather than into the code directory.

A complete G, U, V, J certificate record contains the following information:

schema	format name and version
model	stable model name
input	F, ρ , coefficient field, basis rule, and requested term counts
basis	the ordered exponent pairs used for the unknown and output spaces
matrices	sparse exact records for G and J
reduction	matrix dimensions, selected column vectors $u, v_x, v_y, \alpha, \beta$, and the reduced column
reduced_columns_X	the two-row matrix whose kernel gives A
annihilator_A	derivative form and θ form
certificate_Xi	complete rational Ξ_x, Ξ_y , stored numerators, denominator power, and the exact identity
recurrence	the polynomials Q_k and the indexing convention
constant_terms	independently generated reference values
checks	exact replay results, all required to be true

Matrices and vectors are stored with their dimensions, deterministic labels, declared nonzero counts, and sorted sparse coordinates. Rational expressions are stored after explicit denominator collection and cancellation. Unknown fields are rejected, and a publication record is accepted only when every required proof check is true.

The principal public commands, run from the project root, are

```
python code/run_examples.py public
python code/build_public_release.py Laurent_period_public_release.zip
```

The first command writes its exact report below `examples/public/`; the second uses an explicit allowlist and rejects every private data path. Timing measurements are kept in the root README rather than in the mathematical certificate records.

References

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